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Climate Change and the Financial System

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Abstract

The financial system could help achieve the global climate targets by aligning investments to sustainability. However, investors are largely exposed to carbon-intensive assets that could become stranded, thus delaying the low-carbon transition and bringing new sources of risk for financial stability, i.e., climate-related financial risks. Here, we discuss climate-related financial risks, the challenges they pose to traditional economic and financial risk assessment, and the implications for the implementation and feasibility of climate policies. We then present science-based approaches that introduce forward-looking climate risks and their deep uncertainty in financial risk management (e.g., via the climate value at risk, climate spread, climate stress-test). Finally, we present results of applications aimed at pricing climate risks in investors' portfolios and calculating the largest losses that could lead to systemic risk, in collaboration with leading financial institutions.

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1. INTRODUCTION

At the UNFCCC COP21 meeting that took place in Paris in 2015, most of the United Nations (UN) member countries signed the Paris Agreement aimed at limiting global temperature increase “well below 2°C” above preindustrial levels [Art 2.1(a)] and strengthening the ability of countries to deal with the impacts of climate change [Art 2.1(b)] (UNFCCC 2016). In addition, Art 2.1(c) states the commitment to “making finance flows consistent with a pathway towards low greenhouse gas emissions and climate-resilient development.” Then, in 2016, the UN launched its 2030 Agenda for Sustainable Development, rooted in 17 Sustainable Development Goals (SDGs), while the European Commission launched the EU2030 agenda, which requires the European Union as a whole to reduce its emissions by 40% by 2030.

There is growing awareness of the fact that achieving the global climate and sustainability targets requires aligning finance to sustainability. Indeed, a deep decarbonization of production and consumption activities requires both a massive scaling-up of investments in low-carbon firms and sectors and the phasing out of investments in fossil fuel plants and carbon-intensive sectors. According to the UN's Intergovernmental Panel on Climate Change (IPCC), an annual global investment of US\$2.4 trillion (i.e., 2.5% of the world economy) is needed for retrofitting the energy system alone by 2035 in order to limit temperature increase to 1.5°C from preindustrial levels (IPCC 2018). The magnitude of the investment challenge requires the engagement of public investments, private finance, and capital markets.

However, investments are still largely allocated into economic activities that are not aligned to the climate targets and are at odds with the definition of sustainable finance. The climate stress-test of the financial system showed that investors in the Eurozone and the United States are considerably exposed to financial contracts and securities issued by companies whose revenues are directly or indirectly related to fossil fuels' extraction and combustion (Battiston et al. 2017). Pension funds and investment funds are exposed for circa 40% of their equity portfolios to assets that could become stranded in a disorderly transition to a low-carbon economy. In addition, interlinkages among financial institutions can amplify shocks (both positive and negative) via reverberation of losses in the financial system (Battiston et al. 2017).

It is also increasingly recognized that investors and financial markets are not yet pricing climate-related risks (and opportunities) in the value of financial contracts (Morana & Sbrana 2019, Monasterolo & de Angelis 2020). This means that they might be accumulating (and trading) exposures to risks they are not fully aware of, while they could be dismissing opportunities for returns out of sustainable investment strategies. Mispricing of climate risks could lead to asset price volatility and affect financial stability if large and correlated assets are involved, with potential implications for systemic risk (Monasterolo et al. 2017).

Thus, several central banks and financial supervisors have advocated the introduction of standardized metrics and methods to disclose investors' exposures to climate risks (TCFD 2017) and climate stress-testing exercises to assess the climate-related losses in investors' portfolios that could give rise to macroeconomic and financial risks (IMF 2019, NGFS 2019). In addition, the European Commission has launched its Action Plan on Sustainable Finance (Dombrovskis 2019).

Two main limitations prevent investors and policy makers from embedding climate risk considerations in their investment and policy decisions. First, traditional climate economics and financial risk pricing models are not fit to account for the nature of climate risks (i.e., nonlinearity, deep uncertainty, complexity, and endogeneity) (Battiston & Monasterolo 2019a), as such models are backward looking and consider normal distributions, perfect markets, equilibrium, and lack of arbitrage.

Second, climate economic models that currently produce the climate mitigation and adaptation scenarios and the shared socioeconomic pathways (SSPs) used by the IPCC (2014) and by policy makers do not consider finance and its complexity (Battiston & Monasterolo 2018). This means that they neglect the role of finance as a driver of risk or a barrier to achieving the climate targets and in understanding the impacts of climate change on the financial sector.

In this article, we discuss the nature of climate risks and their main characteristics (forward-looking nature, deep uncertainty, nonlinearity, endogeneity). Then, we present recent advances in climate finance research that developed conceptual and methodological innovations to assess such risks in investors' portfolios. In particular, we discuss the CLIMAFIN tool (Battiston et al. 2019b), a science-based approach to quantitatively assess investors' exposure to forward-looking climate risks, and to price such risks in the value of financial portfolios. By connecting for the first time knowledge developed by climate scientists, climate economists, financial risk and networks experts, and macroecological economists, this new stream of research complements traditional models and helps to inform investment and financial policy decisions.

The review is organized as follows. Section 2 discusses the emergence of climate-related financial risks in the financial and policy debate at the European and international levels. Section 3 analyzes the nature of climate physical and transition risks and the channels through which they can affect economic competitiveness, prices, and financial stability. Section 4 analyzes the limitations of standard economic and financial risk models for the analysis of climate-related financial risks in investors' portfolios, and the implications for policy and financial decision making. Section 5 presents the CLIMAFIN science-based approach to price forward-looking climate risks and their uncertainty in the value of individual financial contracts and investors' portfolios, discussing climate scenario-conditioned financial risk metrics [e.g., climate spread and climate value at risk (VaR)]. Then, Section 6 presents the contribution of the climate stress-test based on financial network models to assess the largest losses induced by climate-related financial risks in investors' portfolios, in light of recent policy applications. Section 7 concludes by discussing the applicability of science-based climate-financial risk metrics and methods to inform investors' risk management strategies and financial supervisors in the identification of systemic climate-related financial risks and in the design of prudential measures to mitigate them.

2. CLIMATE RISKS GAINED MOMENTUM IN THE POLICY DEBATE

Climate-related financial risks have gained attention and popularity among academics, practitioners, and financial institutions since 2015, when the governor of the Bank of England (BoE), Mark Carney, gave a speech entitled "Breaking the Tragedy of the Horizon" at Lloyd's of London, one of the oldest insurance companies in the world (Carney 2015). In his talk, Carney points out that the misalignment between the mid- to long-term realization of climate impacts and the short-termism that characterizes investors' decisions could lead investors to overlook the economic and financial risks stemming from climate change. In 2016, the BoE published a report that formalized a definition of climate physical risk and climate transition risks and the channels through which they could bring material risks for financial actors (Batten et al. 2016).

Other central bankers joined the debate (see, e.g., Signorini 2017, Cœuré 2018). In this regard, standardized metrics and methods for climate-related financial disclosure were advocated by the Financial Stability Board (FSB)'s Task Force on Climate-related Financial Disclosures (TCFD), created by the G20's FSB in its final recommendations (TCFD 2017).

In 2017, a group of central banks and financial regulators created the so-called Network of Central Banks and Supervisors for Greening the Financial System (NGFS) with the aim of

supporting investors to assess climate risks in their portfolios. In 2019, the NGFS included 34 central banks and financial regulators of countries that represent 50% of global emissions and two-thirds of systemically relevant financial institutions. In 2019, the NGFS recommended the introduction of climate stress-tests to assess the financial stability implications of their exposures to climate risks (NGFS 2019).

Some central banks and financial regulators have started to assess investors' exposure to climate risks. The European Insurance and Occupational Pensions Authority (EIOPA), in its "Financial Stability Report" (EIOPA 2018), assessed exposure of the European insurance sector to climate transition risks by building on the methodology developed by Battiston et al. (2017). The same methodology was followed by the European Central Bank (ECB), which assessed Eurozone investors' exposure to economic activities that are considered as "climate policy relevant" in its 2019 "Financial Stability Review" (Giuzio et al. 2019). The Dutch central bank developed climate and energy transition scenario analyses to inform climate stress-testing exercises (Regelink et al. 2017, Vermeulen et al. 2019).

In 2019, a group of central bankers in Europe signed an "Open Letter on Climate-Related Financial Risks" (Bank England 2019) talking about climate-driven "Minsky Moments," i.e., asset price volatility driven by a sudden collapse in prices of fossil-based assets (Carney 2015). In 2020, both the head of the ECB, Christine Lagarde (2020), and the managing director of the International Monetary Fund, Kristalina Georgieva (Ward 2019), recognized the need to assess the exposure of the financial sector to climate-related financial risks and to run climate stress-tests.

Finally, the European Commission, following the recommendations of its High-Level Expert Group on Sustainable Finance (HLEG 2018), launched its Action Plan for Sustainable Finance. In 2019, the European Commission Technical Expert Group on Sustainable Finance published a taxonomy of sustainable assets, green bond standards, and criteria for low-carbon benchmarks aimed at identifying economic and financial activities that contribute to sustainability.

3. CLIMATE CHANGE AND FINANCIAL STABILITY: RISK DRIVERS AND TRANSMISSION CHANNELS

Scientists have recently warned that only a decade is left to implement bold actions aimed at decreasing the impact of human activities on the climate and preventing dangerous and potentially irreversible socioeconomic and environmental impacts (IPCC 2018) and their threats to humanity (Mora et al. 2018). Some of these effects are already visible in the form of more extreme weather events. In this section, we present the notions of climate physical and transition risks, and we analyze their risk transmission channels to the economy and finance.

3.1. Climate Physical Risks

Climate physical risks refer to damages to physical assets, natural capital, and human lives resulting in losses of productive capacity and thus output and gross domestic product (GDP), as a result of climate-induced weather events. Based on the available scientific information (e.g., the IPCC reports), the greenhouse gas (GHG) emissions trajectory currently followed by UN countries would lead to a temperature increase of 3–4°C in comparison to preindustrial levels. This is expected to trigger severe socioeconomic consequences, resulting in particular from sea level rise, ice sheet and permafrost melting, droughts, and floods. The increase in economic damages is already visible (Coronese et al. 2019), with consequences at both the firm and macroeconomic levels:

- The destruction of immobilized productive capital. This has negative implications on firms' profitability, investments, employment, and eventually GDP (Burke et al. 2015, 2018; Hsiang et al. 2017), whereas vulnerable and poor households, in both low- and high-income countries, would suffer the most, being less able to cope with risks (Hallegatte & Rozenberg 2017, Diffenbaugh & Burke 2019). A total of 110 million people live below the current high-tide sea level, and about 250 million people would fall below the level of the worst yearly flood, mostly in low-income countries. This is three times the previous estimates (Kulp & Strauss 2019).
- Drops in property values [see, e.g., the example of luxury coastal properties in Florida and South Carolina that would eventually become not insurable (US Fourth National Climate Assessment Report; <https://nca2018.globalchange.gov/>)]. These have implications for banks and insurance companies. Year 2018 was the fourth-costliest year since 1980 in terms of losses related to natural disasters: Climate-related extremes accounted for €456 billion,¹ i.e., 83 % of the monetary losses over the period 1980–2017 in the European Economic Area (nearly 3 % of its GDP in 2017), and only 35 % circa of these losses were insured, according to the NatCatSERVICE by Munich Re.
- Wildfires. These have become more common worldwide in recent years (e.g., fires raging across eastern Australia and California's Napa Valley in 2019). Climate change can increase the frequency and intensity of wildfires by anticipating spring and summer seasons and by elevating evaporation rates and drought prevalence, thus creating more flammable ecosystems (US Glob. Change Res. Prog. 2018). This adds to human activities in forests, e.g., human encroachment on wooded areas, deforestation (e.g., in Brazilian Amazonia and Indonesia), and questionable forest management. The economic cost of wildfire-induced losses reached US\$5.2 billion (i.e., 0.5 % of global GDP) in Indonesia in 2019 according to the World Bank (Rusmana 2019).
- Water stress (droughts). Climate change has resulted in an increased frequency, intensity, and duration of droughts in some regions (including the Mediterranean, Western Asia, Northeastern Asia, many parts of South America, and much of Africa). Between 1961 and 2013, the annual area of drylands in drought has increased, on average by slightly more than 1 % per year. In 2015, approximately 500 (380–620) million people lived within areas that experienced desertification between the 1980s and 2000s (IPBES 2019).
- Biodiversity loss. The UN's Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services warned that 1 million (and up to 8 million) species are at high risk of extinction due to human activities that affect changes in land and sea use, pollution, and climate (IPBES 2019). Resource depletion and biodiversity loss could affect human well-being well before the tipping point is reached, as already highlighted in 1972 by the *Limits to Growth* publication (Meadows et al. 1972). So far, the loss of pollinators has already put up to US\$577 billion of crop output at risk, while land degradation has reduced the productivity of 23 % of global land (IPBES 2019).
- Soil erosion. This has implications for food commodities' production and prices, the so-called "multiple breadbasket failures" (Janetos et al. 2017). This, in turn, could affect the living conditions of a growing population via price volatility, food production, and consumption patterns (Naqvi & Monasterolo 2019). Consequences include famine and social unrest and, eventually, the relocation of millions of people currently living in areas particularly exposed to climate physical risks, even within developed countries (WEF 2018).

¹This estimate is conservative, insofar as the loss of human life, cultural heritage, or ecosystem services is not part of the estimation.

Most important, the impact of climate change on the processes that withstand desertification and land degradation (e.g., water scarcity or wildfire) may result in compound risks (IPBES 2019) and drive domino effects. Since tipping points in the Earth's system (e.g., loss of the Amazon rainforest or the West Antarctic ice sheet) are interconnected across different biophysical systems, they can potentially commit the world to long-term irreversible changes (Lenton et al. 2019).

3.2. Climate Transition Risks

Climate transition risks refer to the economic and financial losses arising from the revaluation of carbon-intensive and low-carbon assets induced by a sudden change in policy and/or regulation that cannot be fully anticipated by financial actors. Most debated drivers include

- Technological shocks, e.g., the fast decrease of renewable energy production costs and fast increase in their performance, or the change in minimum technology standards;
- Policy and regulatory shocks, e.g., the sudden introduction of a global carbon tax (Monasterolo & Raberto 2018, IMF 2019) or a change in prudential regulation such as the introduction of a green supporting factor (GSF) or a brown penalizing factor that affect capital requirements of financial institutions (HLEG 2018); and
- The sudden changes in “climate sentiments” of financial actors (Dunz et al. 2019) about the implementation of the climate policies, as a result of the coordination of expectations of market participants about the implementation of the climate policies (Battiston 2019).

The adjustments are expected to negatively impact the value of fossil fuel-related assets (the so-called carbon-stranded assets; see, e.g., Leaton 2011). They are also expected to indirectly impact the value of assets in other sectors that use fossil fuel energy and electricity as a production input or that are involved in the value chain of companies that do it, thus generating cascading losses. In contrast, they are expected to positively impact economic activities whose profits directly or indirectly depend upon renewable energy (see Battiston & Monasterolo 2019a for a formalization).

Overall, the notion of climate transition risks stands on the concept of disorderly low-carbon transition (Battiston et al. 2017, NGFS 2019). The transition to a low-carbon economy could occur in an orderly or disorderly way. Orderly means that the climate policies for decarbonizing the economy and achieving the climate targets are introduced early and in a coordinated way among countries. In this context, investors are able to anticipate the climate policy introduction by pricing it in their risk management strategies and thus revising their investment allocations (ideally by decreasing their exposures to carbon-intensive activities and increasing their exposures to low-carbon activities).

In this context, carbon-intensive (low-carbon) firms can adjust their business strategy, avoiding large shocks in their market share, and thus on the value of the financial contracts associated with them (e.g., equity holdings, corporate bonds). Disorderly refers both to the way policies are implemented and to how the market responds to the policies. If climate policies are introduced in a late and sudden way with regard to the decarbonization targets (e.g., EU2030 climate and energy targets), they could trigger a disorderly response from financial actors who may not be able to anticipate (and thus price) the policies in their investment strategies. In this context, carbon-intensive (low-carbon) firms would face unanticipated negative (positive) shocks in their market share, thus affecting the value of the financial contracts issued by such firms.

The lack of investors' anticipation of climate physical and transition shocks could have long-lasting consequences for the economic and financial conditions of a country. On the one hand, if the shock is not anticipated, investors would not change their portfolios' allocations (e.g., by decreasing their exposure to carbon-intensive assets and increasing their exposure to low-carbon

assets). In this context, because asset managers take investment decisions based on the benchmark in their respective markets (Greenwald & Stiglitz 1986), the market benchmark remains carbon intensive, leading to the potential realization of carbon stranded assets in the economy and in finance. Recent research shows that the market benchmark is carbon intensive, as in the cases of the corporate bonds market benchmark and the ECB's corporate bonds purchase [Corporate Sector Purchase Programme (CSPP)] (Battiston & Monasterolo 2019b).

On the other hand, the assessment of the policy shock could be incorrect even on average across market participants, as shown by several recent policy events (e.g., achievement of the Paris Agreement, US withdrawal from the Paris Agreement, outcome of the 2018 Italian elections). These have been incorrectly forecast by most observers and investors but are having severe long-term effects on the financial conditions of a country; see, e.g., the spread on Italy's sovereign bonds (Battiston 2019).

3.3. Climate Risk Transmission Channels: The Role of Policy, Regulation, and Investors' Climate Sentiments

Recent research (Dunz et al. 2019) has formalized the climate risk transmission channels stemming from unanticipated policy and regulatory change and from investors' expectations (climate sentiments) toward the low-carbon transition and assessed their impacts on the credit market and on the performance of agents and sectors of the real economy.

Let us consider the case of the introduction of a GSF, i.e., a decrease in Basel III banks' capital requirements [in the capital adequacy ratio (CAR)] in cases when banks lend to low-carbon firms. The shock hits the bank and then cascades to the carbon-intensive/low-carbon firms via a revision of the lending conditions (worsening for carbon-intensive firms but improving for low-carbon firms) via the interest rate channel (Figure 1). The GSF contributes to decreasing the risk weights for loans to low-carbon firms and to then increasing banks' CAR. Lower interest rates induced by the GSF reduce low-carbon firms' capital costs, which in turn lead to increased aggregate demand. Interest rates are a main channel for investment decisions because they directly affect firms' cost of capital and their unit cost. Indeed, a change in interest rate can affect the relative prices of low-carbon (carbon-intensive) goods, and thus households' budget constraints, and the level and composition of the final demand of the economy. Being more price competitive, the demand for low-carbon capital goods increases and so do the profits for the low-carbon firms, resulting in

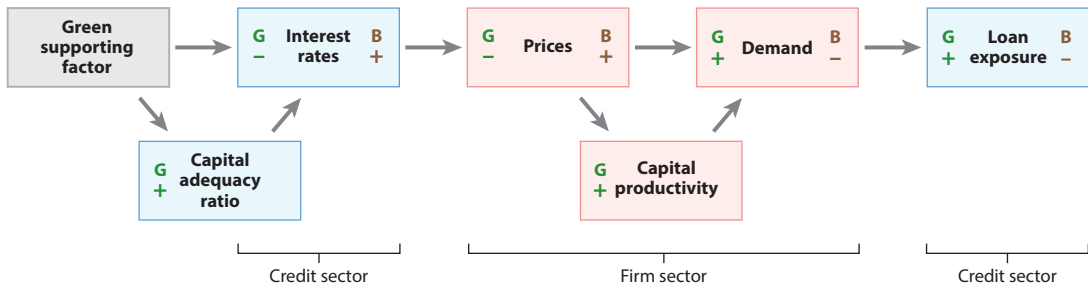


Figure 1

The green supporting factor (GSF)'s risk transmission channels to the credit market and real economy. The grey box represents the policy instrument analyzed. Blue boxes represent the effects of the GSF on the banking sector, whereas pink boxes represent implications for the real economy. Green (G) + (−) operators indicate positive (negative) changes for the low-carbon firms, and brown (B) + (−) operators indicate positive (negative) changes for carbon-intensive firms.

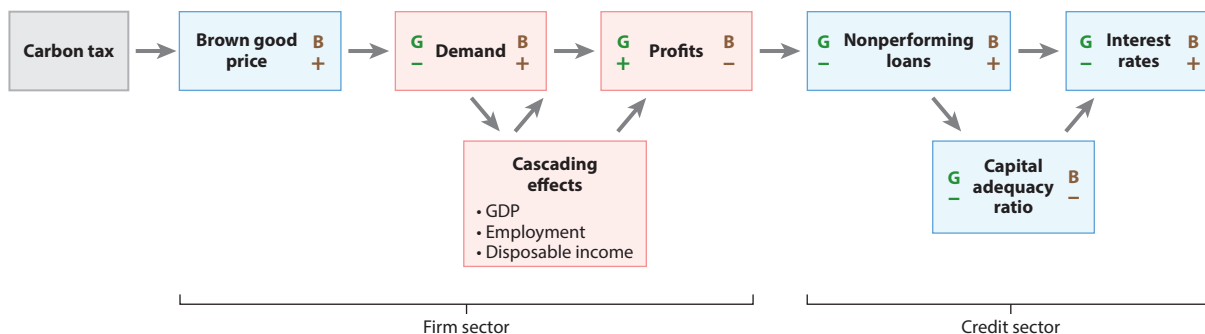


Figure 2

The carbon tax's risk transmission channels to the credit market and real economy. The grey box represents the policy instrument analyzed. Blue boxes represent the effects of the carbon tax on the banking sector, whereas pink boxes represent implications for the real economy. Green (G) + (–) operators indicate positive (negative) changes for the low-carbon firms, and brown (B) + (–) operators indicate positive (negative) changes for carbon-intensive firms. Adapted from Dunz et al. (2019).

higher low-carbon investment and a higher share of green loans for the banking sector (Dunz et al. 2019).

In contrast, in the case of the introduction of a carbon tax, the shock affects the economy via higher production costs for the carbon-intensive firms (**Figure 2**). The cost of the carbon tax can be transferred to households via markup pricing, affecting households' budgets and reducing their demand. In addition, the carbon tax may induce a strong relative price effect in favor of low-carbon capital goods, thus lowering their demand. Both channels contribute to decreasing the profitability of carbon-intensive firms (and thus lower the ability to service the loans) and transfers to the banking sector's lending conditions that become less favorable for such firms in the case of a lower recovery rate (Dunz et al. 2019).

Finally, **Figure 3** shows the risk transmission channels when banks react to the policy announcement by anticipating it in their lending behavior (what we call banks' climate sentiments). The bank does not know the exact timing and magnitude of the policy, but it forms expectations about the future that also accounts for past firm performance and market characteristics. In particular, the bank expects higher (lower) future profits for the low-carbon (carbon-intensive) firms, based on firms' costs and profitability adjustments, and thus adjusts interest rates accordingly.

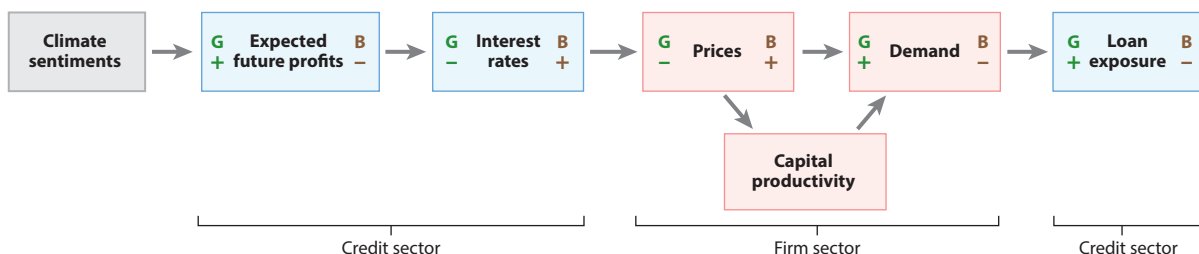


Figure 3

Bank climate sentiments' risk transmission channels to the credit market and real economy. The grey box represents the bank's climate sentiment, i.e., the bank's anticipation of the carbon tax, analyzed. Blue boxes represent the effects of the climate sentiment on the banking sector, whereas pink boxes represent implications for the real economy. Green (G) + (–) operators indicate positive (negative) changes for the low-carbon firms, and brown (B) + (–) operators indicate positive (negative) changes for carbon-intensive firms. Adapted from Dunz et al. (2019).

Lower (higher) interest rates transfer into lower (higher) prices, which in turn have an impact on demand, firms' investments, and then profits in the sectors. As a main difference from the GSF channel, climate sentiments lead the bank to increase interest rates for carbon-intensive firms, thus directly affecting their borrowing conditions. In this context, climate sentiments work as an indirect brown penalizing factor by inducing the bank to anticipate the negative effects of a carbon tax or GSF on the profits of the carbon-intensive firm.

4. CHALLENGES TO INTEGRATE CLIMATE-FINANCIAL RISKS INTO ECONOMIC AND POLICY ASSESSMENTS

So far, the economic impact assessments of climate change produced by integrated assessment models (IAMs) and cost-benefit analyses have neglected the role of finance and its complexity (Battiston & Monasterolo 2018). This may affect the climate policy design and the attainability of the climate targets because the role of finance, as a driver either of climate risks or of climate mitigation, and the impact of climate change on financial risk are not considered. Indeed, climate scenarios that are considered achievable might not be realized owing to the lack of investors' commitment, while other and riskier scenarios could occur. The reason is that the nature of climate risks and their interplay with financial risk introduce several conceptual and methodological challenges for traditional economic and financial models.

4.1. Nature of Climate Risks

Climate change is characterized by the following:

- Nonlinearity of impacts. The probability of forward-looking climate shocks cannot be inferred from historical data, being nonlinear in nature and not normally distributed. For instance, Ackerman (2017) finds that the 2003 Western European summer was 5.4 standard deviations above mean temperature for the period 1864–2000. A similar heatwave occurred in Eastern Europe in 2010. If such events happen every seven years, we cannot assume that temperature changes are normally distributed.
- Deep uncertainties. These characterize impacts and their costs. On the one hand, the deepest climate-related shocks are expected to occur in the mid- to long term, but their exact localization and magnitude are unknown today. This is due to the nature of the Earth system and leads to the presence of tail events (Weitzman 2009), tipping points, and domino effects (Steffen et al. 2018), which are associated with large uncertainty (Kriegler et al. 2009). Tipping points increase the urgency and magnitude of the climate policies to be introduced (Lenton et al. 2019), as well as the cost of inaction for future generations (Lemoine & Traeger 2016). On the other hand, cost-benefit estimates of action (and the cost of inaction) vary substantially across climate scenarios with the assumptions on agents' utility function, future productivity growth rate, and intertemporal discount rate (Stern 2008, Pindyck 2013).
- Forward-looking nature of risk. The impacts of climate change are on the timescale of two decades or longer. However, the time horizon of financial markets is much shorter. Investors' decisions follow a short time horizon (e.g., three months for fund managers) and are based on a market benchmark (performance) that is backward looking because it is estimated on past companies' performance.
- Complexity. The likelihood of the realization of a given climate scenario pathway depends on the ability of countries to introduce coordinated climate policies and on the fact that

socioeconomic agents will react rationally by changing their consumption and production behavior. But because agents often take decisions departing from rational expectations and behaviors, the aggregate effect of heterogeneous agents' behavior can give rise to emerging system's properties that cannot be deduced from the simple sum of individual behaviors (Dosi & Roventini 2019). These properties generate complexity in understanding the system's response to shocks, which can be endogenously generated (Farmer et al. 2015, Battiston et al. 2016), requiring us to think in terms of complex adaptive systems (Lamperti et al. 2019).

- Endogeneity and circularity of climate risk. The likelihood of achieving the global climate targets depends on how climate policies are introduced. Uncertainty about climate policies affects investors' expectations on the financial risk deriving from the very same policies, and thus their investment decision. This generates the possibility of multiple equilibria, a situation where a rational agent cannot identify a preferred investment strategy in the low-carbon transition (Battiston et al. 2017).

4.2. Embedding Climate into Economic and Financial Risk Assessment: Modeling Challenges

It has been increasingly recognized that traditional economic and financial pricing models are not able to embed these characteristics of climate risks (and the opportunities from climate alignment), being constrained by strong assumptions on equilibrium conditions, normal distribution of shocks, linearity of impacts, intertemporal optimization, complete markets, and lack of arbitrage (Balint et al. 2017, Monasterolo et al. 2019). This leads to a potential misunderstanding and underestimation of the impact of climate change in risk assessment models, with relevant implications for policy recommendations (DeFries et al. 2019). Cost-benefit analyses and financial risk valuation build on the identification of the most likely scenarios, in the computation of the expected values and, in the latter case, in the estimation of financial risk metrics that are backward looking in nature because they rely on historical values of market prices (Battiston et al. 2019a).

In contrast, a recent stream of modeling approaches based on complexity economics and network science have provided complementary insights into traditional economic and financial risk models.² In particular, they can enable us to understand three key features of the climate-finance nexus:

- The emergence of reinforcing feedback loops that could give rise to amplification effects, and cascade effects through which risk generating in one sector/agent could be transmitted to others within the network;
- The presence of nonlinearities and tipping points, after which a system changes its core characteristics;
- Policy path dependency, meaning that a measure introduced with a specific goal could have completely different yet long-standing effects on the system.

5. SCIENCE-BASED APPROACHES TO ASSESS CLIMATE-RELATED FINANCIAL RISKS AND IMPACTS

Assessing and pricing forward-looking climate risks and opportunities in the value of financial contracts and investors' portfolios are important to signal investors and the market. Indeed, since

²For a detailed model review, see Balint et al. (2017), Lamperti et al. (2019), and Monasterolo et al. (2019).

financial investors take decisions based on what they can measure, and their decisions influence (and are influenced by) the benchmark in their respective markets, pricing climate risks in financial contracts is crucial from an investors' risk management perspective. In addition, it is important for financial supervisors with a financial stability and supervisory duty mandate because it allows them to assess the onset of instability at the individual company and systemic levels.

Nevertheless, recent research showed that climate risks (and opportunities) are not yet priced in the value of financial contracts and in investor risk management strategies. Karpf & Mandel (2018), Alessi et al. (2019), and Zerbib (2019) assess if a green bond's premium exists, but they find very different results, based on the type of bonds contract analyzed and the "green" definition used. In the catastrophe (CAT) bond market, Morana & Sbrana (2019) find that despite climate-led disasters that have steadily increased from year 2000, the "multiple" (i.e., the return per unity of risk) of the CAT bonds has decreased. Monasterolo & de Angelis (2020) analyze the US, EU, and global stock market reaction to the announcement of the Paris Agreement. They find that the overall systematic risk for the low-carbon indices decreases consistently, while stock markets' reaction is mild for most carbon-intensive indices. Ramelli et al. (2018) analyze the stock market's reaction to the election of Trump as US President and the appointment of the climate skeptic Scott Pruitt as head of the US Environmental Protection Agency and find opposite results, i.e., that investors rewarded companies in high-emissions industries that showed more responsible climate strategies.

5.1. The CLIMAFIN Tool

To fill in this gap, new climate-financial risk metrics and methods have been developed to include the characteristics of climate risks and financial complexity in the assessment of climate economics and policy scenarios and in financial risk evaluation. Battiston et al. (2019b) have developed a transparent, science-based methodology, called the CLIMAFIN tool, that introduces operational metrics and methods to assess climate-related financial risks and impacts.

CLIMAFIN is transdisciplinary and modular and allows

- Classifying investors' exposure in climate policy-relevant sectors (CPRS) based on their direct and indirect contribution to emissions, business line, carbon intensity of revenues, and sensitivity to climate policy (Battiston et al. 2017);
- Pricing forward-looking climate risks in the value of individual financial contracts (equity holdings, sovereign and corporate bonds, and loans) and in investors' probability of default, conditioned to climate scenarios (Battiston & Monasterolo 2019a);
- Introducing forward-looking climate risks into financial risk metrics (such as the climate spread and climate VaR) used by private and public decision makers (Battiston et al. 2017, Monasterolo et al. 2018, Battiston & Monasterolo 2019b); and
- Introducing forward-looking climate risk scenarios and pricing in stress-testing exercises based on financial network models, i.e., the climate stress-test (Battiston et al. 2017), to assess the largest climate-related gains and losses for an investor's portfolio.

CLIMAFIN combines for the first time the climate scenarios developed by climate scientists and reviewed in the IPCC report (e.g., Kriegler et al. 2013, McCollum et al. 2018) with financial risk metrics and methods developed by academics (Battiston et al. 2017) and used by practitioners (EIOPA 2018, Battiston et al. 2019a) in a transparent methodological framework. This allows us to embed forward-looking climate risk scenarios in the valuation of counterparty risk, in the probability of default and largest losses on investors' portfolios.

The methodology is based on the following steps:

1. A definition of the investor information set that considers a risk-averse investor that aims to assess the exposure of her portfolio to forward-looking climate transition risk in a context of incomplete information and deep uncertainty (Knight 1921, Keynes 1973, Nalebuff & Stiglitz 1983, Greenwald & Stiglitz 1986). The investor information set is composed of
 - Climate policy scenarios (provided by, e.g., the IPCC reports);
 - The future climate economic trajectories for carbon-intensive and low-carbon activities, provided by climate models (e.g., IAMs);
 - The definition of forward-looking climate policy shock scenarios; and
 - The assessment of the shocks on sectors' gross value added (GVA) and firms' profitability conditioned to forward-looking climate policy shock scenarios.
2. A definition of the investor's risk management strategy, e.g., minimizing the worst-case losses of the portfolios under different forward-looking climate scenarios. The definition of the risk management strategy should account for an investor's specific risk aversion levels, for counterparty risk adjusted for climate policy shock scenarios (e.g., probability of default, spread), and for metrics relevant for financial regulation (e.g., a risk measure such as VaR).
3. A definition of the investor's portfolio of risky (defaultable) financial contracts and securities, which could include equity holding, corporate bonds, sovereign bonds, and loans.
4. Assessment of the macroeconomic and climate transition risk trajectories at the issuer and counterparty levels. The impact of future climate shocks (e.g., the sudden introduction of a carbon tax in 2030 that agents do not fully anticipate) is calculated at the level of output of economic activities in carbon-intensive and low-carbon sectors, in their contribution to the countries' GVA and fiscal revenues, and in the change in firms' cash flows and profitability.
5. A valuation model to price equity risk and credit risk conditioned to forward-looking climate transition risk using the climate policy shock scenarios (from the investor information set). This requires embedding climate nonlinearity and ambiguity in the analysis of how a forward-looking climate shock shifts the probability of default of a contract and cumulatively on the whole portfolio.
6. The definition of climate spread and climate VaR. The climate spread is defined as the change in spread of a corporate or sovereign bond contract conditional to a given climate policy shock scenario, thus introducing future climate risks in the assessment of firms or countries' financial solvability. The climate VaR can be defined as the "worst-case loss" under future climate shock scenarios given a certain confidence level.

5.2. Pricing Climate Transition Risks in a Sovereign Bond Portfolio: The Case of Central Banks

Battiston & Monasterolo (2019a) assess the exposure of sovereign bonds issued by OECD countries (10 years, zero coupon) and include in the portfolio of the Austrian central bank (OeNB) to forward-looking climate transition risk scenarios derived by IAMs (**Table 1**). They consider the impact of a tight climate policy scenario, based on the sudden introduction of carbon pricing aiming to keep global temperature increase below 2°C in comparison to preindustrial levels (StrPol-450) and on the change in output and market share of low-carbon and carbon-intensive sectors using two IAMs: WITCH and GCAM (Kriegler et al. 2013). Then, they model the impact of the change in low-carbon and carbon-intensive sectors' profitability on the change in the GVA and fiscal revenues of the OECD country of residence. Finally, they price the shock on fiscal revenues in the value of the sovereign bonds of the issuing country, and the sovereign bonds' yields, i.e., the climate spread. They find that the largest negative shocks on the value of individual

Table 1 Climate policy shock on OECD's sovereign bond value and climate spread^a

Region	Model region	WITCH: bond shock (%)	WITCH: yield shock (%)	GCAM: bond shock (%)	GCAM: yield shock (%)
Austria	EUROPE	1.3	−0.16	0.13	−0.02
Australia	REST_WORLD	−17.36	2.45	NA	NA
Belgium	EUROPE	0.84	−0.1	0.03	0
Canada	PAC_OECD	−5.21	0.67	−18.29	2.61
Norway	REST_WORLD	−14.82	2.05	NA	NA
Poland	EUROPE	−12.85	1.75	−2.49	0.32

^aNote that positive shocks on the yield correspond to negative shocks on the value of the sovereign bond. Climate spread: 2.45 = 245 basis points. Data from Battiston & Monasterolo (2019a).

sovereign bonds are associated with countries (e.g., Australia and Norway) where fossil fuel-based primary and secondary energy sources represent a large contribution to the GVA and thus on the countries' GDP. In contrast, the sovereign bonds of countries (e.g., Austria) with growing shares of renewable energy sources on the GVA experience positive shocks. This means that accounting for the level of (mis)alignment of a country's economy to the low-carbon transition, under feasible climate transition scenarios, could already affect its financial risk position. In particular, the level of climate alignment of a country's economy could already be reflected in the value of its sovereign bond and change it considerably.

5.3. Climate Risk Assessment of the Portfolios of Development Finance Institutions

The CLIMAFIN approach to the climate-financial risk exposure of investors' portfolios has been applied to the analysis of development banks' project loans, conditioned to climate physical and transition risk scenarios. Monasterolo & Battiston (2016) assess the exposure of the Caribbean Development Bank (CDB)'s top 100 projects (in capital value, with a 30-year maturity) in the Caribbean region, across all areas of economic activities (e.g., agriculture, infrastructure, energy, manufacturing). They find that ca. 60% were highly exposed to climate physical and transition risks, and ca. 47% contributed to increasing the islands' climate vulnerability by supporting investments in fossil fuel developments, gas pipelines, and construction in areas previously hit by natural disasters and yet built without an adaptation strategy. In contrast, only a minor share of the portfolio (12%) contributed to increasing climate resilience through adaptation measures (e.g., in infrastructure developments) (**Figure 4**). This result points out the misalignment of the current investment strategy of the bank to the climate and sustainability objectives and the challenges for its ability to deliver on its mandate as well as to preserve its financial solvability; for example, if Caribbean countries default, like in Grenada in 2004–2006 after Hurricane Ivan, the recovery rate on the loans is close to zero.

In 2018, using the CLIMAFIN methodology to assess investors' exposure to CPRS, Monasterolo et al. (2018) develop a carbon risk exposure of Chinese development banks' portfolios to overseas energy infrastructure projects. They analyze 199 overseas energy investment loans (from oil-based primary energy to solar-based electricity production) provided by two main Chinese policy banks (China Development Bank and Export-Import Bank of China). The loans covered 63 low- and mid-income countries from 2000 to 2018, for a value of US\$228.105 billion. They find that the exposure of both banks to losses driven by climate transition risk on their overseas energy infrastructure loans is mostly driven by coal- and oil-based energy projects in East Asia. The largest negative shocks range between 4% and 22% of their portfolios, meaning that

		Exposure to climate risks				
Contribution to adaptation		NA	H	M	L	Total
	Contribution to adaptation	0	0	1.52	10.57	12.09
	Contribution to vulnerability	0	13.33	33.43	0	46.77
	Not classifiable	33	0.51	6.63	0.94	41.15
	Total	33	18.84	41.59	11.51	100

Figure 4

The Caribbean Development Bank's portfolio exposure (top 100 projects in capital value) to climate risks and contribution to adaptation in 2016. Numbers are in percent. The red cell represents the share of projects at the highest climate-financial risk because of their high exposure to climate risks and high contribution to vulnerability. The yellow cell represents the share of projects at an average climate-financial risk because of their medium exposure to climate risks and medium contribution to vulnerability. The bright green cell represents the share of projects that carry the lowest climate-financial risk because of their low exposure to climate risks and high contribution to adaptation. The light green cell represents instead the share of projects that carry medium climate-financial risk because of their average exposure to climate risks and medium contribution to adaptation. Adapted from Monasterolo & Battiston (2016). Abbreviations: H, high; L, low; M, medium; NA, not available.

with an average leverage of 12, even an average shock (ca. 10%) could lead the banks to financial distress (Table 2).

5.4. Climate Risk Assessment of the Sovereign Bond Portfolio of European Insurers

More recently, the CLIMAFIN framework was applied to provide a forward-looking climate transition risk assessment of the sovereign bond portfolios of solo insurance companies in Europe, as a result of the first collaboration between climate economists, climate-financial risk modelers,

Table 2 Exposure of Chinese policy banks' overseas energy portfolio to climate transition risk in low-income and mid-income countries. Adapted from Monasterolo et al. (2018)

Model	Scenario	MinShock	MaxShock	TotNegShock	TotPosShock	TotRelNegShock	ProjectVar
GCAM	LIMITS-RefPol-450	−1,582	6,500	−12,176	22,338	−0.05	−830
WITCH	LIMITS-RefPol-450	−3,965	6,500	−50,778	42,809	−0.22	−3,878
GCAM	LIMITS-RefPol-500	−1,180	6,500	−9,484	21,957	−0.04	−711
WITCH	LIMITS-RefPol-500	−3,503	6,500	−31,343	46,723	−0.14	−2,710
GCAM	LIMITS-StrPol-450	−1,585	6,500	−13,275	23,742	−0.06	−1,001
WITCH	LIMITS-StrPol-450	−3,852	6,500	−49,281	42,954	−0.22	−3,587
GCAM	LIMITS-StrPol-500	−1,192	6,500	−10,838	23,360	−0.05	−893
WITCH	LIMITS-StrPol-500	−3,482	6,500	−31,139	46,745	−0.14	−2,721

and financial regulators (Battiston et al. 2019a). The analysis considers forward-looking scenarios characterized by a disorderly introduction of climate policies (i.e., carbon pricing) that cannot be fully anticipated and priced in by investors. The authors analyze the shock on the market share and profitability of carbon-intensive and low-carbon activities in the economy to which insurers are exposed via sovereign bonds, conditioned to climate transition risk trajectories provided by the LIMITS IAMs. Then, they define the climate risk management strategy under uncertainty for a risk-averse investor (insurer) who aims to minimize the largest losses in her sovereign bond portfolio. The analysis allows pricing the climate transition scenarios in the probability of default of the individual sovereign bonds and in the bonds' climate spread. Then, after revaluating the financial contracts, the authors estimate the largest gains/losses on the insurance companies' portfolios. They find that the potential impact of a disorderly transition to a low-carbon economy on insurers' portfolios of sovereign bonds is moderate but non-negligible under several feasible scenarios.

6. FINANCIAL VALUATION IN NETWORK MODELS FOR CLIMATE-FINANCIAL RISK ANALYSIS

In the last decade, and in particular after the 2008 financial crisis, network models have been increasingly developed by scholars to assess the role of financial interconnectedness and complexity on the emergence of systemic risk (Battiston et al. 2012). Financial valuation in network models provided a tool to capture second-round effects (i.e., loss amplification) in stress-test and contagion exercises of central banks and financial regulators, and thus to identify the onset of financial instability. Battiston & Martinez-Jaramillo (2018) identify the contribution of network models to financial risk and credit valuation in the ability to

- Assess systemic risk in case of the relevant impact of indirect exposures on losses, diversification of risk across counterparties and external assets, and interconnectedness;
- Understand the impact of externalities on financial contracts and in the buildup to systemic risk, considering the presence of imperfect information and incomplete risk markets; and
- Estimate the feedback effects between the financial system and the real economy, where the network effects can better help to explain macroeconomic aggregate phenomena.

6.1. From Stress-Tests to the Climate Stress-Test

Recently, network models have been developed and applied for stress-testing portfolios of financial institutions against forward-looking climate risks, with the aim of identifying the drivers of the largest losses that could have systemic effects. In this context, financial valuation in network models offers a framework to integrate climate risk consideration into financial and credit risk assessment. In particular, it allows the analysis of individual investors' direct and indirect exposures to climate risks at the level of individual financial contract and security and to assess the largest losses on investors' portfolios. The advantages of using financial network models include the possibility to account for the three main drivers of systemic risks in financial networks:

- Financial interconnectedness that could amplify both positive and negative climate shocks and significantly decrease the accuracy of estimations of default probabilities, increasing the complexity of risk estimation (Battiston et al. 2012);
- The leverage of individual financial institutions and endogenous recovery rate in the calculation of the resilience to shocks (Barucca et al. 2020); and

- The possibility of mispricing individual financial contracts that are traded across investors in the financial network, where risk can reverberate (D'Errico et al. 2018).

The climate stress-test of the financial system by Battiston et al. (2017) provides an application of financial valuation in network models to the analysis of the exposure of individual investors to equity holdings issued by economic activities that are relevant for climate transition risk, i.e., the so-called CPRS. These include fossil fuels, utilities, energy-intensive sectors, housing, and infrastructure and are identified as based on the direct and indirect contribution of economic activities (classified at the NACE 4-digit level) on GHG emissions; their relevance for climate policy implementation (e.g., the EU carbon leakage directive 2003/87/EC); the technology mix of the activity, including its capital expenditures (Capex); and the role of the business in the value chain (Battiston et al. 2017).

With the climate stress-test, the authors assess the first- and second-round effects, i.e., the indirect losses due to devaluation of counterparties' debt obligations on the interbank market, using the DebtRank algorithm (Battiston et al. 2012). Finally, they calculate the climate VaR conditioned to different climate transition risk scenarios provided by the LIMITS IAMs (Kriegler et al. 2013) and to low- or high-carbon investment strategies of the 20 most exposed banks in the European Union and United States (**Figure 5**). The VaR allows the measurement (in dollar amount) of the potential loss in value of a risky asset (e.g., equity) or portfolio over a defined period for a given confidence interval, and thus it is considered as a measure of risk of investment loss.

The authors find that the exposure of individual investors to CPRS is large, heterogeneous across agents, and amplified by network effects. In particular, the exposures of pension funds and investment funds reach 43% and 45% of their equity portfolios, respectively, and the potential losses could be amplified by the mutual exposures of financial actors (e.g., pension funds and investment funds). Roncoroni et al. (2019) further develop the climate stress-test in an application

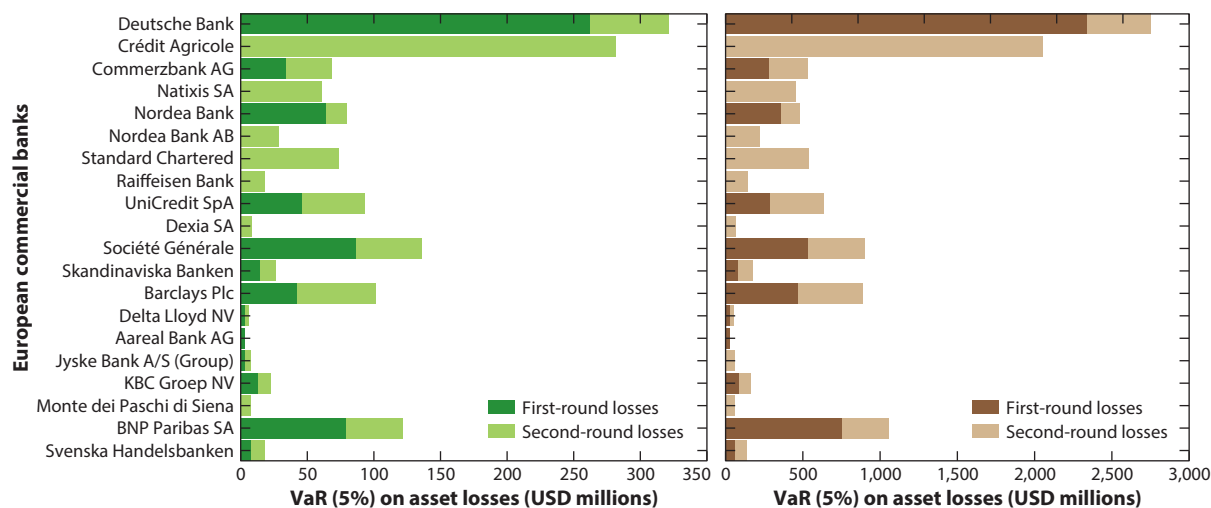


Figure 5

Climate value at risk (VaR, 5% significance) on equity holdings of the 20 most severely affected banks, under a scenario of renewable (green) and fossil (brown) fuel investment strategies. The results can be interpreted as follows. Relative to a baseline of no climate policy introduction, banks that aligned their investment strategy to the low-carbon transition targets earlier (i.e., those that started to divest from climate policy-relevant sectors and invest in the low-carbon sector) would benefit from positive volatility on assets prices. In contrast, investors who continued with the business-as-usual portfolio allocation could face losses in asset value. Interestingly, banks' exposure to shocks via first and second rounds is highly heterogeneous. Adapted from Battiston et al. (2017).

to Banco de Mexico's portfolio, where the asset network valuation framework approach (NEVA) (Barucca et al. 2020) is extended to a network of banks and investment funds and includes endogenous recovery rate.

6.2. Financial Macro-Network Models for Climate Policy Analysis

Financial macro-network models (Castrén & Rancan 2014) have been further developed and applied to assess the drivers and channels of climate risk transmission to financial actors and to socioeconomic agents and sectors, accounting for reinforcing feedback loops and cascading effects. Stolbova et al. (2018) assess the impact of a disorderly transition in reaction to the introduction of the most debated climate policies (i.e., carbon tax, green quantitative easing, and a GSF) on the balance sheet of individual financial institutions in the Eurozone (i.e., banks, investment funds, pension funds, insurance) through direct and indirect chains of financial exposures across multiple financial instruments (equity, loans, bonds). The financial macro-network model allows consideration of the role of financial interconnectedness and the closed chain of contracts, presence of bankruptcy costs, not fully collateralized debt contracts, and a recovery rate lower than one to provide a comprehensive assessment of climate-financial cascading losses from financial to economic agents.

Importantly, the model proposed by Stolbova et al. (2018) does not include a behavioral dimension and thus does not allow capturing the effects of the policy on the expansion of low-carbon sectors. Nevertheless, recent developments in financial macro-network behavioral models (Naqvi & Monasterolo 2019), which integrate features of multilayer networks and agent-based models, can capture the change in agents' investment behavior and the resulting emergent change in macroeconomic aggregates in response to a climate shock.

7. CONCLUSION

Climate risk differs from other sources of risk because it is forward looking and global (yet with differentiated impacts) and characterized by deep uncertainty, nonlinearity, and endogeneity. In contrast, in economics and finance, investment decisions with long-lasting societal impacts are typically taken based on backward-looking benchmarks and short time horizons. Standard economic and financial risk assessment approaches are inadequate to deal with the complexity of climate risk in the economy and society. This has important implications for the design of policies and investment strategies, and thus for the feasibility to achieve the climate targets. In particular, missing the interplay between climate and financial risks could prevent the direction of climate finance flows toward those segments of the economy and society that will be more affected but less prepared to build resilience.

In this article, we discussed why climate change is now considered as a source of risk for the financial system, and we introduced the metrics and methods developed by academics and used by financial institutions to assess climate risks and opportunities in investment and policy decisions. In particular, we presented the main conceptual and methodological innovations introduced by the CLIMAFIN tool (Battiston et al. 2019b), which for the first time integrates knowledge produced by different yet interconnected fields of research (i.e., climate economics, financial risk pricing, network models, and macroecological economics) in financial risk valuation. First, CLIMAFIN provides models to price forward-looking climate economic trajectories in the value of individual financial contracts and securities (equity, sovereign and corporate bonds, and loans) under deep uncertainty. Second, it introduces forward-looking climate transition risks in traditional financial risk metrics such as the (climate) spread and the (climate) VaR of an investor's portfolio. Third,

it integrates the results of the climate risk pricing models and climate-financial risk metrics in a climate stress-test framework, rooted in network models. This allows estimating the largest losses for individual investors' portfolios conditioned to disorderly climate transition scenarios.

The results obtained so far show that climate-financial risk can materialize soon for several types of investors, changing private and public financial risk positions, based on both the exposure of investors' portfolios to CPRS and the composition of the economy and sovereign debt. On the one hand, the relevance of carbon-intensive (low-carbon) activities on GVA can negatively (positively) affect firms' profitability, impacting on sovereign fiscal revenues and on the probability of default of sovereign bonds, via a change in the climate spread. This means that the level of climate (mis)alignment of a country's economy could already be reflected in the value of its sovereign and affect its refinancing conditions on the international market. In addition, under several climate scenarios, the potential impact of a disorderly transition to low-carbon economy on financial actors (e.g., pension funds, investment funds and insurers, development banks) is considerable, and thus deserves to be regularly monitored and assessed.

On the other hand, financial interconnectedness and the characteristics of the financial network could amplify shocks (both positive and negative) via reverberation of gains and losses. This, in turn, could lead to financial instability with implications on systemic financial risk. Further, research results show that private and public investment decisions can have a large impact on the decarbonization of the economy and thus on the achievement of the climate targets.

The climate and financial risk metrics and methods presented here are now under review or already being applied by several financial institutions, central banks, and financial regulators with a financial stability or sustainable development mandate. For these institutions, understanding under which conditions finance could be a driver of or a barrier to climate (mis)alignment, and to what extent climate scenarios could affect financial stability, is fundamental to deliver on their mandate.

Given the large and growing role that the financial sector plays in the economy and society, and the long-lasting effects of financial crises, the assessment of climate-related financial risks should be considered as a public good and should be mainstreamed in economic and financial assessments and in policy evaluation. In this regard, transparent, science-based metrics and methods are crucial for risk management strategies of investors' portfolios in the low-carbon transition for financial supervisors to identify the drivers of systemic risk in financial system, and to design measures to mitigate such risks and preserve financial stability.

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